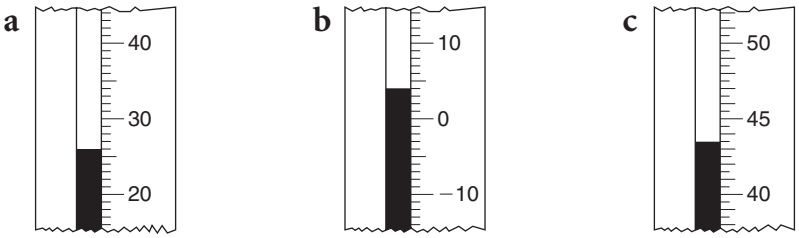
What's the temperature?

1 In the middle of July, if the temperature outside were 20 °C you might comment, 'It's quite cool today.' However, in January if the temperature outside were 20 °C you might say, 'It's unusually warm today.'

Explain why you would describe the same temperature so differently.

2 The diagram shows the mercury level in some thermometers. What temperature is each thermometer showing in degrees Celsius?



3 **a** Suggest three reasons why mercury is a good liquid to use in thermometers.

b Suggest one disadvantage of using mercury in a thermometer.

4 Why would you use an alcohol thermometer rather than a mercury thermometer to measure the temperature in a freezer?

5 Here are some temperatures.

–10 °C	**0 °C**	**10 °C**	**20 °C**	**29 °C**
37 °C	**58 °C**	**98 °C**	**100 °C**	**110 °C**

Which of these temperatures is:

a the temperature of melting ice?

b normal body temperature?

c the temperature of boiling water?

d the temperature in a freezer?

e air temperature on a hot summer day?

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Warming up and cooling down

1 The table shows the results for a cooling experiment. The water in two beakers A and B was allowed to cool and the temperature was recorded every minute.

Time (minutes)	Temperature of water in beaker A (°C)	Temperature of water in beaker B (°C)
0	80	80
1	75	70
2	69	62
3	65	56
4	61	51
5	58	48
6	54	44
7	52	42

a Plot a graph of temperature (*y*-axis) against time (*x*-axis) for the water in beaker A. Draw the best-fit curve and label it A.

b On the same axes plot the points for temperature against time for beaker B. Draw the best-fit curve and label it B.

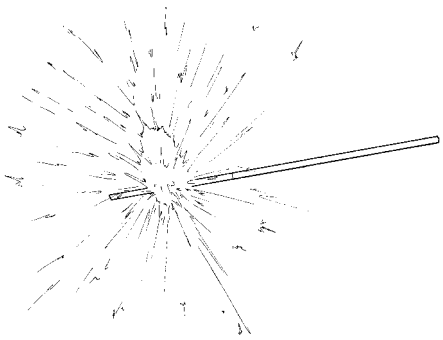
c In which beaker did the water cool more quickly? Explain how you can see this on the graph.

d The two beakers were the same size and contained the same volume of water. Suggest why the water in one beaker cooled more quickly than the water in the other beaker.

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Temperature and heat energy

1 The sparks from a sparkler at a Bonfire Night party are very hot. Each spark is a tiny piece of white-hot iron. If these white-hot sparks hit your clothes, they do not burn the clothing.



Explain why something so hot does not burn your clothes.

2 List the following objects in order, in two different ways:

a in temperature order – highest temperature first

b in heat energy order – the one that you think contains most heat energy first.

Objects

a beaker containing 100 cm^3 of water at 100 °C
a beaker containing 500 cm^3 of water at 90 °C
an ocean with an average temperature of 5 °C
a 100 cm^3 lump of copper at 95 °C

3 If you placed a 1 kg block of ice at 0 °C into a bucket of water at room temperature (20 °C), what might you expect the temperature of the water to be:

a after 5 minutes?

b the next day?

Choose from these answers:

–10 °C	**0 °C**	**10 °C**	**20 °C**	**30 °C**

4 Two identical electric kettles are switched on at the same time. Kettle A contains 1 litre of water, and kettle B contains 0.5 litres of water. In both kettles the temperature of the water is 20 °C. The water in kettle A takes longer to reach boiling point than the water in kettle B. Why is this?

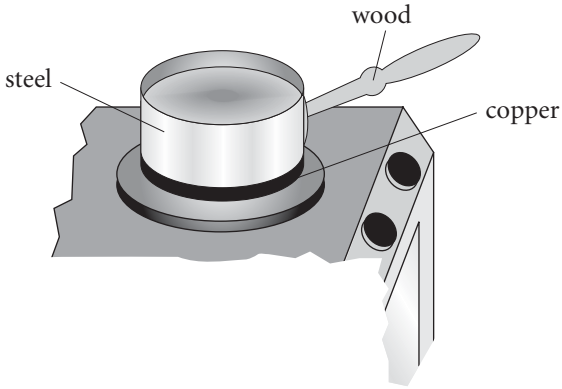
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What are conductors and insulators?

1 A metal rod is heated at one end. The heat travels along the rod by conduction. Explain, in terms of particles, how the process of conduction occurs.

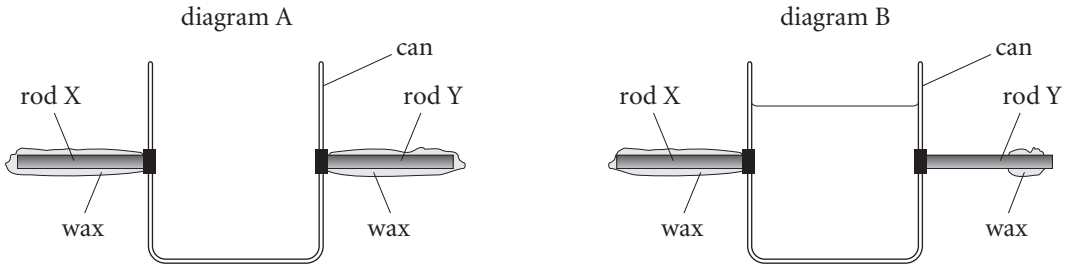
2 A saucepan is made mainly of steel, but it has a copper bottom and a handle made of wood.

Explain the reasons for the use of each of these three materials.



3 On a cold day, the steel handlebars on a bicycle feel colder than the rubber handlebar grips. Explain why this is.

4 Diagram A shows two metal rods, X and Y, that are made of different metals. They are the same length and thickness. They are coated with wax and the ends are fixed through the walls of a can. Hot water is poured into the can and the wax begins to melt. After a few minutes the apparatus is as shown in diagram B.



a Explain why the wax melts.

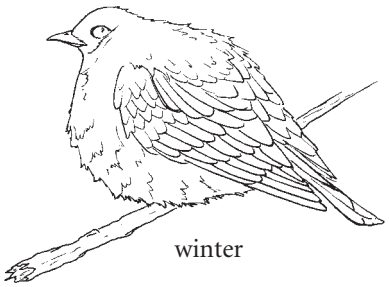
b Explain why more wax melts on rod Y than on rod X.

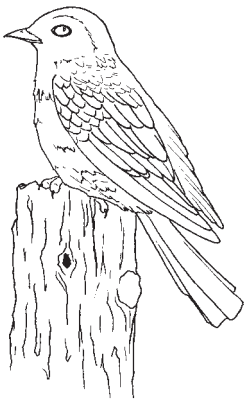
5 Rewrite the following list of materials in order of thermal conductivity. Write the best conductor first and the poorest conductor last.

aluminium	**copper**	**ice**	**polystyrene**	**wood**

6 In winter, birds fluff up their feathers to keep warm.

Explain why fluffing up the feathers helps birds to keep warm.

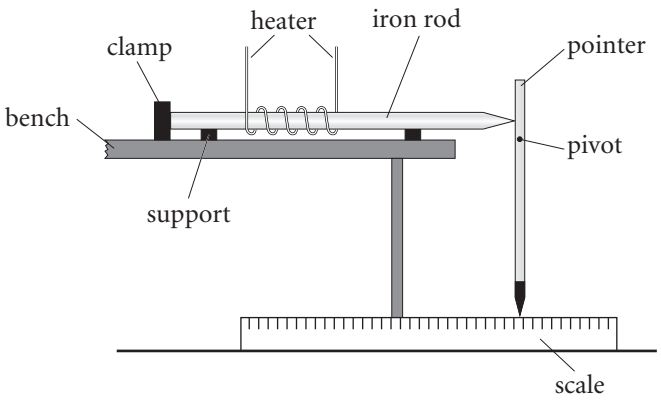




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Expanding and contracting

1 The diagram shows apparatus used for demonstrating that a metal rod expands when heated.



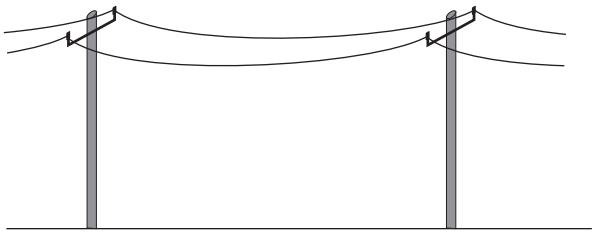
a Explain why the pointer moves along the scale when the rod is heated.

b Why is it necessary to have a long pointer?

c Explain what happens to the pointer when the rod is allowed to cool.

2 The diagram shows some telephone wires that have been put up in warm weather.

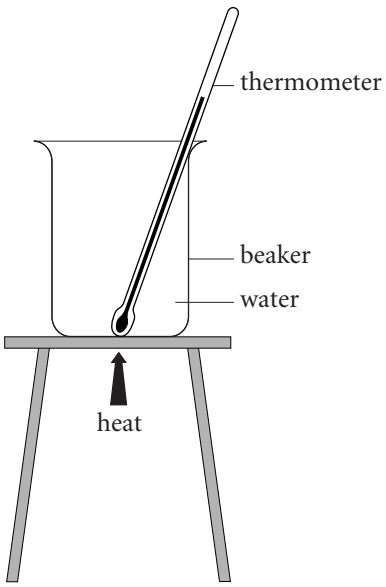
Why is it important that the wires have been left slack?



3 Explain why a metal screw cap on a glass jar can often be unscrewed more easily after being warmed in hot water.

4 Explain why a milk bottle would probably crack if boiling water were poured into it.

5 If you were heating some cold water in a beaker to a temperature of about 60 °C, why would it not be sensible to fill the beaker to the top with cold water?

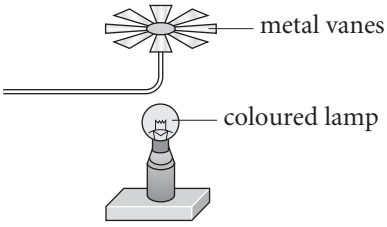


6 When a gas canister used for a camping stove has 'run out', it does, in fact, contain gas at atmospheric pressure. No more gas comes out because to do so its pressure has to be greater than atmospheric pressure.

Explain why it would be extremely dangerous to throw the canister on to a fire.

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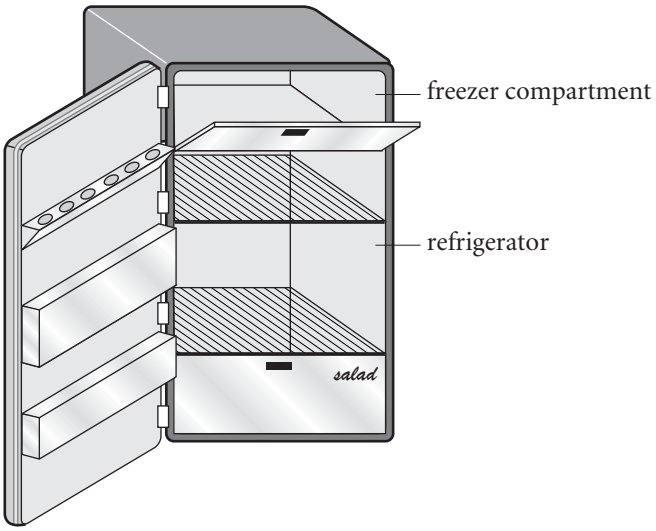
Convection



1 The flickering effect on many electric fires is produced by a set of vanes pivoted above a coloured lamp.

Explain how this makes the flickering effect.

2 The freezer compartment inside a refrigerator is at the top of the cabinet.

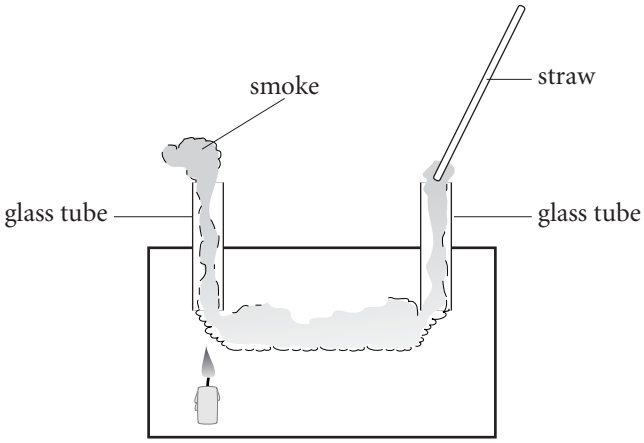


Explain, in terms of convection:

a why it is at the top

b why the refrigerator would be ineffective if the freezing compartment were at the bottom.

3 The diagram shows a model mine. The smoke represents fresh air. The candle represents a fire.



Use your knowledge of convection to explain how the fire produced a flow of fresh air in the mine for the miners to breathe even though they were deep in the mine.

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Radiation and energy escape

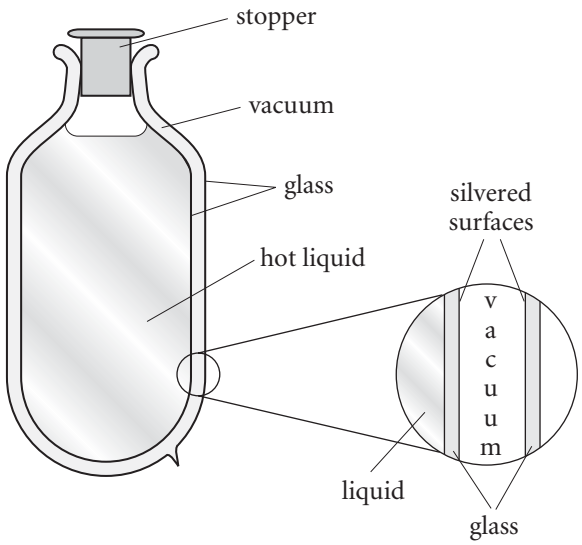
1 A vacuum flask is designed to reduce heat transfer by conduction, convection and infrared radiation.
Which parts of the vacuum flask are designed to reduce heat transfer by:

a conduction?

b convection?

c infrared radiation?

Briefly explain your answer in each case.



2 A vacuum flask is usually used to keep liquids hot. However, it can also be used to keep a cold liquid cool.

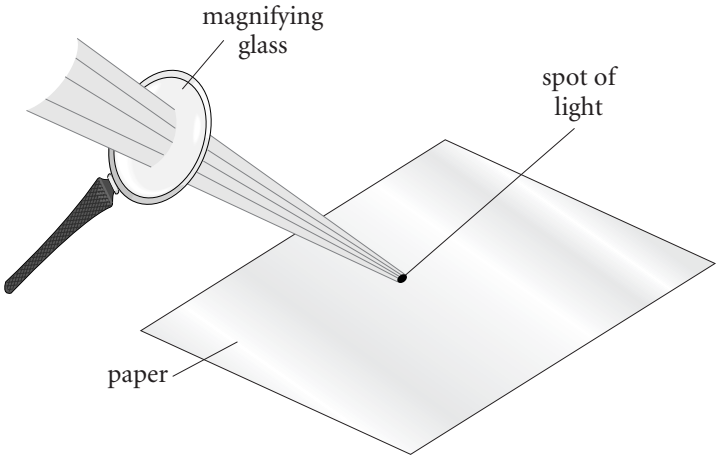
a Explain how it works in this case.

b Why would it not be sensible to have ice cubes in the cold liquid in the flask?

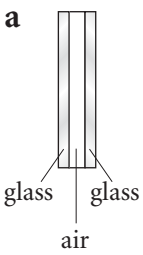
3 A magnifying glass can be used to focus light from the Sun on to a piece of paper. At the point where the light is focused, the paper may char or burn.

a Explain the similarity between light and infrared radiation that this shows.

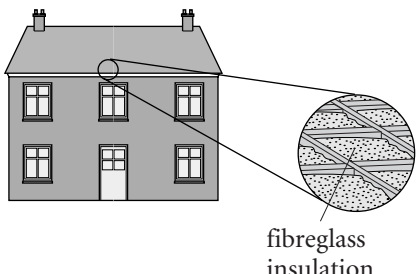
b State two other similarities between light and infrared radiation.



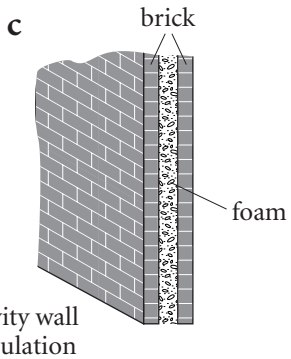
4 The diagrams show three ways that heat losses from homes can be reduced.



double glazing



roof insulation



cavity wall insulation

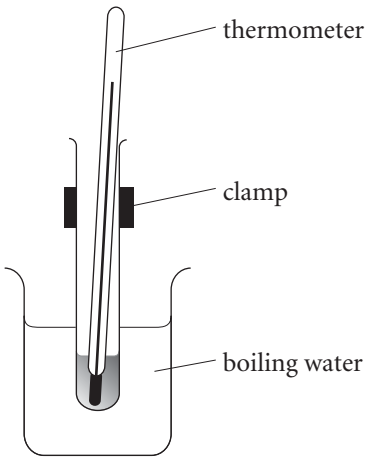
Explain how each of them helps to prevent heat losses.

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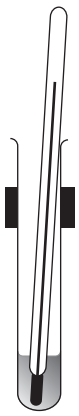
Explaining changes of state

1 The diagram shows the apparatus used in an experiment to record the temperature as a substance cools. The substance was heated until it was all molten, and then it was allowed to cool. The temperature was taken as the substance cooled. The readings are shown in the table.

A heating



B cooling



Time (minutes)	0	2	4	6	8	10	12	14	16	18
Temperature (°C)	110	90	70	70	70	50	40	30	20	20

a Use the data in the table to plot a graph of temperature (*y*-axis) against time (*x*-axis).

b Use your graph to answer these questions:

i What is the melting point of the substance?

Choose from: **110 °C** **90 °C** **70 °C** **50 °C** **20 °C**

ii What was room temperature?

iii What was the temperature at 3 minutes?

c Describe the state of the substance in the tube at the following times.

Choose from: **solid** **liquid** **gas**

i after 2 minutes

ii after 5 minutes

iii after 10 minutes

d Copy and complete the following passage by choosing the correct words from the list below.

melting **heat energy** **constant** **slowing** **closer** **temperature** **level** **boiling** **particles**

As the substance cools, it gives out energy and its _____ drops. At the _____ point the temperature stays _____. The _____ in the substance are _____ down and moving _____ together. While this is happening, _____ is released which stops the substance from cooling down.

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